

DAKSH DHANWANI

Indore, Madhya Pradesh, India

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LinkedIn

GitHub

Education

Shri G. S. Institute of Technology and Science (SGSITS), Indore

2023 – 2027

Bachelor of Technology in Electronics and Instrumentation | CGPA: 7.70

Indore, Madhya Pradesh

Experience

Reliance Industries Limited

May 2026 – July 2026

G.E.T. Summer Intern

Navi Mumbai, Maharashtra

- Reviewed **P&IDs**, control loops, and instrumentation architecture across gas turbines, boilers, and utility units to understand field instrumentation and process control systems in a live industrial environment.
- Engineered a unified **SAP Interchangeability List** by conducting physical hardware audits of **ABB 800xA DCS** and **Rockwell PLCs**, streamlining CAPEX/OPEX procurement pipelines and reducing potential maintenance downtime.
- Analyzed and mapped complex control architectures for a 240 MW Captive Co-generation Power Plant, focusing on **Triple Modular Redundancy (TMR)** in **GE Frame 6B Mark VIe** controllers and **3-Element Drum Level** boiler logic.

DA-IICT, Gandhinagar

May 2025 – July 2025

Research Intern

Gandhinagar, Gujarat

- Built an **embedded sensing system** on Raspberry Pi 3B integrating temperature, humidity, and pressure sensors for real-time data acquisition and on-device processing.
- Developed an end-to-end **Edge AI pipeline** for fully offline inference, deploying a lightweight transformer-based model on Raspberry Pi 3B for generating natural-language summaries from live sensor data.
- Collected, preprocessed, and evaluated **real-time sensor datasets** while optimizing the model for resource-constrained embedded deployment.

Projects

DCS Valve Control Simulator — ABB 800xA-Inspired Architecture | HTML, JavaScript, Chart.js

July 2026

- Built an interactive simulation of a **DCS-based valve control loop**, modelling the signal path from HMI setpoint through I/O, **4–20 mA** transmission, I/P conversion, and pneumatic control-valve actuation with PID feedback and real-time SP–PV trending.
- Simulated instrumentation failure scenarios including **wire breaks** and **instrument-air loss**, with corresponding fail-safe valve responses and basic OT network segmentation based on industrial control concepts studied during the RIL internship.

Real-Time Environmental Monitoring System | Python, Raspberry Pi

July 2025

- Developed a standalone embedded monitoring system using **Raspberry Pi 3B**, DHT11, and BMP280 sensors for continuous temperature, humidity, and pressure acquisition, local processing, and real-time monitoring without cloud dependency.

Technical Skills

Instrumentation & Control: P&ID Interpretation, Control Loops, DCS Monitoring, 4–20 mA Current Loops, Instrument Calibration

Embedded & Hardware: Raspberry Pi, Arduino, Sensor Interfacing, I²C, UART, Microcontrollers, Edge Computing, Digital Circuit Design

Programming: C++, Python, SQL

Tools: Git, Linux, VS Code, Jupyter Notebook, MS Office (Excel, Word)

Relevant Coursework

Industrial Instrumentation, Sensors & Transducers, Control Systems, Process Instrumentation, Electronic Measurements, Analog & Digital Electronics, Microprocessors & Microcontrollers

Leadership Experience

Placement Coordinator

June 2024 – Present

Shri G. S. Institute of Technology and Science (SGSITS), Indore

- Coordinated campus recruitment activities for **500+ students**, managing recruiter communication, candidate registrations, scheduling, and on-ground execution across placement drives.
- Served as a point of contact between recruiters, the Training & Placement Cell, and candidates, resolving operational issues and ensuring timely execution of recruitment activities.